DIALibGen: predicted spectral libraries for DIA proteomics

October 2026 — DIALibGen 0.12.1

What it does, how to use it, and how it compares with DIA-NN.

Summary

DIALibGen predicts DIA spectral libraries (retention time, fragment intensities, ion mobility) from a protein FASTA and can fine-tune its retention-time and ion-mobility predictors on one of your runs. It is a single command-line program, with a desktop app, for Linux, macOS and Windows.

Measured against DIA-NN 2.0's built-in predictor on one Orbitrap Astral dataset and a whole-proteome library:

	DIALibGen	DIA-NN 2.0
library from a FASTA, 5.8 M precursors, 96 CPU threads	146 s (0.11.3)	172 s
the same at 16 CPU threads	683 s (0.11.3)	458 s
the same on 2 × H100 GPUs, 16 CPU threads	48 s (0.12.1)	– (not measured)
median |ΔRT| (min), 16,068 held-out precursors, LOWESS-calibrated; untuned → fine-tuned on the same training peptides (0.11.1)	0.243 → **0.093**	0.290 → 0.142 [1]
median cosine to observed y-ion intensities, 40,419 precursors, ions stored by both libraries (0.11.1)	**0.986**	0.977
precursors passing DIA-NN 2.0's estimated 1 % FDR, run 01, untuned → fine-tuned library (0.11.1; the tuned libraries were trained on this run)	89,232 → **96,156**	88,130 → 90,843

[1] A second DIA-NN tune, trained on DIA-NN's own search results (more training precursors, validation proteins included), reached 0.099 min (section 4).

On a MacBook Pro (Apple M4 Max, 16 cores) DIALibGen 0.11.4 generates the same 5.8 M-precursor library in 444 s on the CPU and in 298 s with the fragment model on the Apple GPU (258 s once its compiled shapes are cached); the Astral library is fine-tuned in 71 s with GPU training, the default on Apple silicon (section 3). The server runtime rows above were measured with DIALibGen 0.11.3 on a non-public human-plus-entrapment FASTA (5.81 M precursors); 0.11.4 and 0.11.5 keep the CPU generation path and write the same library content. Each tool's time includes its own fragment selection and output format. The accuracy and identification rows use libraries made with 0.11.1 and 0.11.3 on the public human FASTA (3.49 M precursors); 0.11.4 generates the same CPU library content (checked for the generated Astral library and one CPU tune; tuned libraries depend on the training setup). DIALibGen also predicts and can tune CCS/ion mobility; this report evaluates only RT, fragment intensities and identification counts. The CPU timings for the two tools come from two identical machines (section 3). The accuracy tables use run 01 of the ProteoBench Astral DIA dataset (section 4); no other instrument was tested.

1. What DIALibGen is

DIALibGen digests a protein FASTA in silico and predicts, for every precursor, the retention time (RT), the fragment-ion intensities and, for instruments that separate ions by mobility (timsTOF), the collisional cross section (CCS), from which it derives the inverse reduced ion mobility (1/K0, in V·s/cm²); since 0.12 a library for an instrument without ion mobility (Astral, other Orbitraps) carries no CCS or 1/K0. Prediction uses the open AlphaPeptDeep models, run through ONNX Runtime from C++; digestion, modifications and file handling use OpenMS. Prediction runs on the CPU or on NVIDIA GPUs. DIA-NN 2.0 predicts libraries itself; DIALibGen is a standalone, scriptable predictor that also writes OpenSWATH formats, records how each library was made, and can fine-tune its RT and CCS models on your own run.

Release packages exist for Linux (x64, arm64), macOS (Intel, Apple silicon) and Windows (x64): a command-line archive and a desktop app that drives the same program. The prediction models are bundled. Since 0.11.4 the macOS release uses the Apple GPU: fine-tuning trains there by default on Apple silicon (`–machine:device auto` resolves to Metal, MPS); since 0.12.1 `auto` also predicts the fragments there (CoreML) for libraries of at least 50,000 precursors per CPU thread, and `–machine:device coreml` does so for any size (RT and CCS stay on the CPU, which CoreML cannot replace for these models). `–mode prepare` compiles the GPU shapes of the fragment model once, ahead of a first run. Since 0.12.1 the Linux packages include ONNX Runtime's CUDA build, so prediction runs on an NVIDIA GPU when the CUDA 12 runtime and cuDNN 9 are installed, and on the CPU otherwise.

Modes

Every RT in a library must be on a single scale, because a search engine calibrates the library as a whole. DIALibGen libraries carry RT either on the model's normalised 0–1 scale (freshly predicted; optionally rescaled to the indexed RT scale, iRT, with `–generation:irt_rescale true`) or in the minutes of one run (observed or tuned values). The mode rules below exist to prevent mixing the two. Ion mobility (1/K0) can also be predicted or observed, but has no scale problem.

mode	input	output
`generate` (default for a FASTA)	protein FASTA	predicted library: RT, fragment intensities (and, with ion mobility, CCS and 1/K0) for every precursor of the digest
`refine`	library + identifications from one run	observed RT (and, with `–write_im`, 1/K0) written into the library; by default only the identified precursors are kept
`tune`	library + identifications from one run	the whole library re-predicted with RT and/or CCS models fine-tuned on that run; with the RT model tuned, RT in that run's minutes
`convert` (default for a library)	library	the same library in another format
`prepare` (macOS)	–	the fragment model's CoreML shapes compiled into the cache, for later `–machine:device coreml` runs

generate needs no run data. It digests the FASTA, enumerates the precursors (charges, modifications, m/z range), predicts RT, fragment intensities (b/y ions, charge 1–2 by default) and, for an instrument with ion mobility, CCS, derives 1/K0 from the CCS (by default), and writes the library.

refine writes observed values; it does not tune by default.

1. Read the DIA-NN report of one run (a multi-run report is refused).

2. Keep the rows that pass the q-value thresholds (an estimated 1 % false discovery rate, FDR), by default 0.01 each for the precursor, global and protein q-values (a threshold of 1 turns that gate off). An enabled gate needs its column; a missing column is an error. Decoys and rows with an invalid q-value or charge are dropped.
3. Match the rows to library precursors by modified peptide sequence (UniMod spelling normalised) and charge.
4. By default keep only the matched, identified precursors.
5. Replace their RT with the observed RT. Optionally (`–write_im`) replace 1/K0 as well, for charges of at least `–im_min_charge` (2); the CCS derived from the old 1/K0 is then blanked. If the instrument's mobility ramp top is given (`–im_ramp_top`), observations within `–im_ramp_margin` (0.02) of it are treated as missing and the prediction stays. Optionally (`–write_intensity`) copy in observed fragment intensities; this needs a report with fragment quantities (`––report–lib–info` or `––export–quant`) and applies a fragment-quality gate.
6. `–out_report` writes the residuals measured *before* the overwrite; after it they would be zero by construction.

Two cases would mix the scales: keeping unidentified precursors (`–no_filter`), or an observed RT for only some of the identified ones. Both are refused unless RT is left alone (`–no_write_rt`) or the library is first re-predicted in the run's minutes by `–tune –tune_heads rt|both`, which runs the tune procedure, then step 5. CCS-only tuning does not fix the RT scale and is refused there. A filtered library also changes the next search: precursors missed in the first run stay missed, and the error-rate estimate refers to that smaller set.

tune adapts the models; it writes no observed values and removes nothing. Defaults in brackets.

1. Read the report (one run; protein groups are needed), keep rows with q ≤ `–filter:q_value` (0.01), drop decoys, rows without a protein group and, for CCS, charges below `–filter:min_charge` (2).
2. Build training units: one per modified sequence for RT, one per sequence and charge for CCS; the observed value is the median over the unit's rows. For RT, a sequence whose rows spread by more than `–filter:rt_spread_max` (0.2 min) is dropped. The RT target is the observed RT divided by the largest observed RT among the units, or by `–filter:rt_max_minutes` if that option is set, so targets lie in 0–1; the CCS target is computed from the observed 1/K0, m/z and charge.
3. Split by protein group with a hash: about one group in five goes to a test set, one in seven of the rest to a validation set, the remainder is the training pool (the shares in sequences come out as 20 % and 13 % in section 4). Each of test and validation needs at least 100 units, otherwise the tune is refused (`–cohort:` options change this split).
4. Fine-tune the pretrained AlphaPeptDeep RT (or CCS) network: Adam, peak learning rate 1e-4 with a linear warm-up and cosine decay, mean absolute error as the loss, mini-batches of 1,024, gradient clipping, up to `–train:epochs` (100) passes over the training units; the LSTM initial states stay frozen.
5. After each epoch (`–stop:eval_every` 1), score the validation set: by default the standard deviation of the residuals after a straight-line calibration. A checkpoint is kept only if it beats the pretrained ("stock") model on validation; if none does, the tune is refused. Training stops after `–stop:patience` (10) epochs without an improvement of at least `–stop:rel_tol` (0.5 %) over the best so far, never before epoch 20, or earlier at the epoch or time budget.
6. Report the test-set metrics; under the default cohorts the test set never took part in selection.
7. Re-predict the **whole** library with the tuned model(s), identified and unidentified precursors alike. With the RT model: RT = model output × the denominator of step 2, so the library is in the run's minutes. Fragment intensities, precursor keys and m/z are unchanged.
8. Record the recipe (cohorts, settings, seed, thread count) in the Parquet output or the `.refine.json` sidecar of a TSV, and in the tuned ONNX files, which are kept only with `–tune_out_models`. CPU training uses `–machine:threads` (16 on Linux and Windows, 8 on macOS); the documentation states that the tuned weights may differ between thread counts (on our Mac, 8 and 16 gave identical weights). Training on a GPU (CUDA in a source build, or Metal on Apple silicon) is not bit-identical to the CPU.

On data without ion mobility (Astral, Orbitrap), pass `–tune_heads rt`: the default `both` needs observed 1/K0 in the report.

convert writes an existing library in another format without changing its content. Every mode can write DIALibGen Parquet (one row per precursor, with the recipe and input checksums embedded), DIA-NN TSV (read by DIA-NN 1.7 and 2.x), DIA-NN 2.0's Parquet layout (`-out_format diann2_parquet`) and OpenSWATH PQP or TSV (`-out_format openswath_tsv`). DIA-NN reads the TSV and its own Parquet layout, not DIALibGen's Parquet.

Where the identifications come from. `refine` and `tune` take one run's identifications from a DIA-NN report (`-ids report.parquet`); this is the validated route and the one used throughout this report. Alternatively, `-run` gives them the run itself (centroided mzML or mzPeak): DIALibGen then searches the run, writes its identifications as a report and continues exactly as with `-ids`. `-run` is not a mode of its own, and it is experimental: its validation is not complete (error-rate control checked with entrapment sequences, i.e. sequences that cannot be in the sample and so expose false identifications; and resource limits). With `-run`, `-write_intensity` is not available; the CCS head and `-write_im` need a diaPASEF run converted with its per-peak 1/K0 values and searched with ion mobility; the intended form is `-mode refine -tune -no_filter`.

A typical workflow:

1. `generate` a library from the FASTA (DIA-NN TSV, or `-out_format diann2_parquet`);
2. search one representative run with DIA-NN and that library;
3. `refine` with that search's report (a library carrying the observed RTs, an "empirical" library), or `tune` (an adapted predicted library), or `refine -tune -no_filter` for both; `tune` picks RT, plus CCS when the report has ion mobility (`-tune_heads auto`, the default since 0.12);
4. search the other runs with the result.

With DIA-NN 2.x, for an Astral run (DIALibGen 0.12.1; all physical cores by default):

```
DIALibGen -in proteins.fasta -out predicted.tsv -generation:instrument Astral
diann --f run01.raw --lib predicted.tsv --fasta proteins.fasta --reannotate \
  --met-excision --out first_pass.parquet --threads 16
DIALibGen -mode tune -in predicted.tsv -ids first_pass.parquet -out tuned.tsv
```

Choose `-generation:instrument` for your instrument (the Astral results here use `Astral`, which selects AlphaPeptDeep's Lumos-group fragment model). Since 0.12.1 the instrument's file of defaults also sets the NCE, the m/z windows and the charges: for `Astral` NCE 25, precursor m/z 380–980, fragment m/z 150–2000 and charges 1–4, the settings of this report's Astral libraries; for `timsTOF` charges 1–4 and m/z 350–1200 / 200–1800; other instruments use the generic defaults (350–1200 / 200–1800, charges 1–4). Match digestion, modifications, charges and m/z ranges to the experiment. Modifications are given by name (OpenMS or UniMod), one per residue; this report evaluates only carbamidomethylated cysteine and oxidised methionine, and accuracy for other instrument settings or modifications is not established here. A whole human proteome takes 5–10 GB of RAM to generate and 6–7 GB to tune (section 3); its OpenSWATH PQP export needs about 60 GB.

Installation, the command lines of the benchmark libraries (in their 0.12.1 form), their settings and the thread and GPU options are in Appendix A.

2. Data

FASTA (sections 4 and 5). UniProt human reference proteome UP000005640, one sequence per gene (`UP000005640_9606.fasta`, 20,652 entries: 20,329 Swiss-Prot, 323 TrEMBL), UniProt release 2026_03, downloaded on 9 October 2026 from https://ftp.uniprot.org/pub/databases/uniprot/current_release/knowledgebase/reference_proteomes/Eukaryota/UP000005640/UP000005640_9606.fasta.gz (proteome page https://www.uniprot.org/proteomes/UP000005640). md5 of the uncompressed file `503f8006bb0a3aa0e002031d5fb5ce49`. The `current_release` link moves with every UniProt release; older releases are under https://ftp.uniprot.org/pub/databases/uniprot/previous_releases/.

Runs. The three runs of the "DIA Ion Entrapment – Astral" module of ProteoBench, a community benchmarking platform for proteomics software: `LFQ_Astral_DIA_15min_50ng_Human_01`, `_02`, `_03`: technical replicates of a human digest, 50 ng, 15-min gradient (the acquisition lasts about 24 min; the identified peptides elute between 3 and 22.5 min), Orbitrap Astral in DIA mode. Module documentation: https://proteobench.readthedocs.io/en/latest/modules/dia/entrapment-dia-astral/; raw files: https://proteobench.cubimed.rub.de/raws/DIA-astral-entrapment/all_data_Entrapment_DIA_Astral.tar.gz (5.6 GB). The vendor `.raw` files were converted to mzML with OpenMS `FileConverter` (Thermo RAW reader, development build of 9 October 2026), keeping the instrument's centroided peaks. The module's own pre-digested entrapment FASTA was not used here; all searches used the human FASTA above. The files have no PRIDE accession that we could identify.

FASTA for the runtime comparison (section 3). A proteome-sized FASTA from an earlier entrapment study, 40,862 entries: 20,431 reviewed human UniProt sequences (February 2021, with common contaminants and the Biognosys iRT peptides) and 20,431 matched entrapment sequences (md5 `3fb427c3514e897a2d53e3f28b25967e`). It is not public, so the runtime benchmark cannot be repeated independently; it serves only as a large input of human-like sequences (for scale: the public FASTA above gives a 3.49 M-precursor library in 121.7 s on the same node, section 3). No timsTOF run is used in this report.

Software versions. Six DIALibGen versions produced the results; their libraries are the same where the table says so. The command lines are written for 0.12.1, the release this report accompanies; of 0.12.1 only the server GPU runtime was measured.

version	used for	note
DIALibGen 0.11.1 (release build)	library set A: accuracy and DIA-NN 2.0 identifications (sections 4, 5)	its untuned library is identical to 0.11.3's and 0.11.4's (CPU)
DIALibGen 0.11.2	runtime rows that bridge the two servers (section 3)	same library as 0.11.3
DIALibGen 0.11.3 (source build)	server runtime (section 3); library set B: the DIA-NN 1.7.12 identifications and the entrapment check (section 5, Appendix A)	0.11.4 generates the same CPU library content
DIALibGen 0.11.4 (Homebrew cask)	laptop runs, CPU and Apple GPU (section 3)	CPU libraries equal the server's to float rounding; CoreML libraries differ slightly in fragment intensities
DIALibGen 0.11.5 (source build, CUDA)	server GPU runtime, one H100 (section 3)	its GPU library is byte-identical to 0.11.3's
DIALibGen 0.12.1 (release package, Linux x64)	server GPU runtime, one and two H100 (section 3); the command lines (section 1, Appendices A and B)	its GPU library equals 0.11.3's (Parquet table; only the recorded recipe differs); its Astral (Appendix A) and timsTOF (section 3) commands write the same TSV as 0.11's longer commands
DIA-NN 2.0 (Academia, 28 Jan 2025)	predictor runtime; set A tuning, searches and RT reference	
DIA-NN 1.7.12 (open-source Linux build)	set B training identifications and searches; intensity reference	cannot predict libraries

3. Runtime against DIA-NN

Task. Predict a library from the 40,862-entry FASTA with identical digest settings in both tools: Trypsin/P (DIA-NN: `--cut K*,R*`), 1 missed cleavage, length 7–30, charge 2–4, precursor m/z 350–1200, fragment m/z 200–1800, fixed Carbamidomethyl, no variable modifications, Met excision. Both write 5,808,438 precursors. DIALibGen keeps up to 12 fragments for each (69,699,513 transitions; Parquet 766 MB) and predicts RT, fragments and CCS/1/K0 with the `timsTOF` instrument setting, which makes both tools predict ion mobility as well, at normalised collision energy (NCE) 30; the instrument label only conditions the fragment model and does not change the cost. DIA-NN 2.0 (Academia build of 28 January 2025) predicts spectra, RT and ion mobility. The commands (`F` = the FASTA, `N` = threads):

```
diann --fasta F --fasta-search --predictor --gen-spec-lib --cut 'K*,R*' \
  --missed-cleavages 1 --min-pep-len 7 --max-pep-len 30 --min-pr-charge 2 \
  --max-pr-charge 4 --min-pr-mz 350 --max-pr-mz 1200 --min-fr-mz 200 \
  --max-fr-mz 1800 --unimod4 --met-excision --var-mods 0 --threads N \
  --out-lib lib.parquet

# DIALibGen 0.12.1 (timstof.yaml supplies NCE 30 and the m/z windows)
DIALibGen -in F -out lib.parquet -generation:instrument timsTOF \
  -generation:precursor_charges 2 3 4 -generation:min_relative_intensity 0 -threads N

# the runs here (0.11.x) used the same settings as a recipe, which 0.12.1 also reads:
# recipe.json: {"precursor_charges":[2,3,4],"instrument":"timsTOF",
#               "min_relative_intensity":0.0,"nce":30.0}
DIALibGen -in F -config recipe.json -out lib.parquet -threads N
```

Machines. `kim` and `dax` are identical servers (2 × AMD EPYC 9654, 2 × 96 cores, 384 hardware threads, Linux); `data` and `spock` have the same CPU model (224 hardware threads online) and 2 × NVIDIA H100 PCIe each. The laptop is a MacBook Pro with an Apple M4 Max (12 performance and 4 efficiency cores, 40-core GPU, 128 GB, macOS 26.6.2), plugged in, the 1-minute load below 3 before every run (a low load average did not mean an idle machine: a background indexer used about one core during several runs); it ran DIALibGen 0.11.4 from the Homebrew cask (CoreML for the fragment model, Metal for training; the RT and CCS models stay on the CPU because CoreML has no operator for their LSTM layers). A first 0.11.4 CPU generate that ran next to a busy 3D viewer (826.1 s) was discarded and repeated; both libraries are byte-identical. Server CPU timings were not repeated with 0.11.4 or 0.11.5: the CPU generation path is unchanged. The one-H100 rows of 0.11.5 were measured on `spock` with 0.11.3 as a control in the same session, on GPU 1 with 16 CPU threads of its own socket (another user's job held most of GPU 0's memory); the first 0.11.5 run spent 10.1 s loading its libraries before the program started (cold file cache) and was repeated. The 0.12.1 rows (the published Linux x64 package, which ships ONNX Runtime's CUDA build, with cuDNN 9 on the library path; the same recipe as above) were measured on `spock` on 16 October 2026 with 0.11.3 as a control in the same session: one H100 on GPU 1 with CPUs of its own socket, as for 0.11.5, and two H100 with both GPUs idle; the first 0.11.3 control on one H100 spent 9.2 s loading its libraries before the program started and was repeated. Five more 0.12.1 runs on two H100 took 47.9–48.8 s. Each configuration ran once (0.11.2 at 96 threads on `kim` three times), measured with `/usr/bin/time -v` (on the laptop `/usr/bin/time -l`), and started only when the node was quiet (fewer than 24 hardware threads busy on `kim`/`dax`; load below 0.25 per hardware thread and the GPUs used idle and holding no other user's job on `data` and `spock`). DIALibGen's times include the collection of a timing trace; its overhead was not measured. DIALibGen and DIA-NN do not write the same library: DIALibGen keeps 12 fragments per precursor and its own Parquet layout, DIA-NN its own fragment selection (not counted here) and layout. The times are therefore end-to-end times for each tool's library, not for identical prediction work.

Table 1: Library generation from the FASTA, 5.8 M precursors

tool	node	threads	wall time (s)	peak RAM (GiB)	CPU time (s)
DIA-NN 2.0	kim	8	860.6	6.8	6,530
DIA-NN 2.0	kim	16	458.5	7.1	6,567
DIA-NN 2.0	kim	96	171.9	16.4	10,262
DIALibGen 0.11.2	kim	8	1,155.9	4.2	9,193
DIALibGen 0.11.2	kim	16	595.1	4.7	9,252
DIALibGen 0.11.2	kim	96	158.2 [1]	9.5	12,309
DIALibGen 0.11.2	dax	16	593.6	4.8	9,240
DIALibGen 0.11.2	dax	96	157.9	9.5	12,310
DIALibGen 0.11.3	dax	16	682.8	4.9	10,863
DIALibGen 0.11.3	dax	96	146.2	9.7	12,532
DIALibGen 0.11.2	data	16 + 1 × H100	88.9	–	–
DIALibGen 0.11.2	data	16 + 2 × H100	65.2	–	–
DIALibGen 0.11.3	spock	16 + 1 × H100	79.2	4.6	–
DIALibGen 0.11.5	spock	16 + 1 × H100	79.1	4.7	–
DIALibGen 0.11.3	data	16 + 2 × H100	47.3	5.0	–
DIALibGen 0.11.3 [3]	spock	16 + 1 × H100	79.6	4.7	–
DIALibGen 0.12.1 [3]	spock	16 + 1 × H100	77.7	4.6	–
DIALibGen 0.11.3 [3]	spock	16 + 2 × H100	48.6	5.1	–
DIALibGen 0.12.1 [3]	spock	16 + 2 × H100	48.3	5.0	–
DIALibGen 0.11.4	laptop	16 (CPU)	443.6	6.0	6,532
DIALibGen 0.11.4	laptop	16 + CoreML, cold [2]	298.5	7.7	1,067
DIALibGen 0.11.4	laptop	16 + CoreML, warm [2]	258.3	7.9	1,486

[1] mean of three runs (158.10 / 158.22 / 158.24 s); every other row is a single run. CPU time = user + system. Peak RAM = peak resident memory on the host (GPU memory not included). DIALibGen 0.11.3 was not run at 8 threads. [2] MS2 model on the Apple GPU, RT and CCS on the CPU at the same time; cold: the compiled-model cache was emptied before the run, so it includes compiling the model's shapes for the GPU; warm: the shapes are loaded from the cache. [3] one session, 16 October 2026.

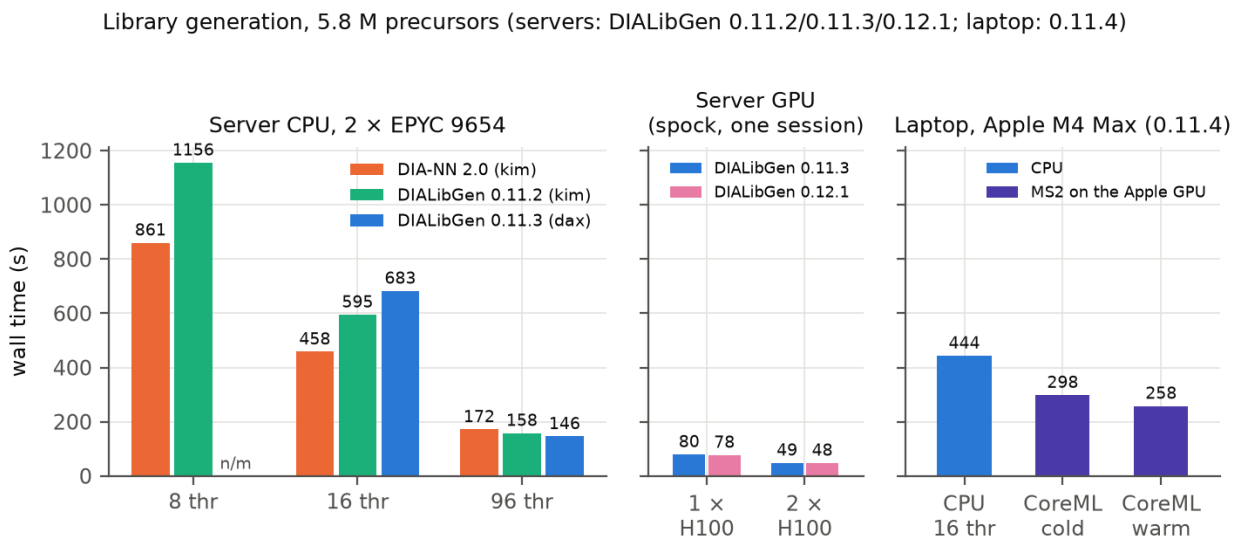


Figure 1: Library generation from the FASTA, 5.8 M precursors, wall time. Left: the two identical servers, CPU only. Middle: `spock`, 16 CPU threads plus one or two H100, DIALibGen 0.12.1 (release package) and 0.11.3 in the same session. Right: the laptop (Apple M4 Max, DIALibGen 0.11.4), CPU at 16 threads and with the fragment model on the Apple GPU (CoreML; cold and warm compiled-model cache). Each configuration was measured once, except DIALibGen 0.11.2 at 96 threads on kim (mean of three); the two tools' libraries differ in fragment selection and layout. "n/m": not measured.

Reading the table:

- DIA-NN 2.0 and DIALibGen 0.11.3 were not timed on the same node. DIALibGen 0.11.2 was timed on both nodes and the times agreed within 0.3 % (595.1 vs 593.6 s at 16 threads, 158.2 vs 157.9 s at 96). This supports comparing across the two nodes, but it is two measurements, not a measured error bound.
- **At 96 threads** DIALibGen 0.11.3 took 146.2 s, DIA-NN 2.0 171.9 s (DIA-NN 1.18 × longer), one run each.
- **At 8 and 16 threads DIA-NN 2.0 is faster**: 458.5 s against 682.8 s for 0.11.3 at 16 threads (1.49 ×), and 860.6 s against 1,155.9 s for 0.11.2 at 8 threads. For libraries of the same 5,808,438 precursors, DIALibGen uses 1.2–1.7 × DIA-NN's CPU time. 0.11.3 is 15 % slower than 0.11.2 at 16 threads and used 17 % more CPU time there (10,863 vs 9,252 s): 0.11.2 computed only the fragment-model outputs that are written, 0.11.3 computes the model's full output (same weights, neutral-loss channels included, which are never written). At 96 threads the CPU times differ by 2 % and 0.11.3 is the faster one, because other parts of the run dominate there; the libraries are identical.
- DIA-NN 2.0 needs more memory at every thread count: 6.8–7.1 GiB at 8–16 threads (DIALibGen 4.2–4.9 GiB) and 16.4 GiB at 96 threads (DIALibGen 9.5–9.7 GiB).
- During the DIA-NN runs, other users' jobs kept about 18 of the node's 384 hardware threads busy on average; during the DIALibGen runs, below 1 on `kim` and 1.6–1.9 on `dax`. The effect on DIA-NN's times was not measured.
- **GPU server (0.12.1, 0.11.5).** The 0.12.1 release package takes 77.7 s on one H100 and 48.3 s on two, against 79.6 s and 48.6 s for 0.11.3 in the same session; its library equals 0.11.3's apart from the recorded recipe. On one H100 0.11.5 took 79.1 s, 0.11.3 79.2 s in its own session; those libraries are byte-identical. 0.11.4 was slower on NVIDIA GPUs (89.2 s on one H100 and 57.1 s on two, against 79.3 s and 48.6 s for 0.11.3): it predicted retention time and ion mobility at the same time as the fragments, which helps on the Apple GPU but made the three models share an NVIDIA GPU. 0.11.5 runs them one after the other there again.
- **Laptop (0.11.4).** The 16-core M4 Max generates the 5.8 M-precursor library on the CPU in 443.6 s, faster than the server at 16 threads (682.8 s) with 40 % less CPU time. With the fragment model

on the Apple GPU it takes 298.5 s with an empty compiled-model cache and 258.3 s once the shapes are cached; in 0.11.4 the RT and CCS phases run on the CPU while the GPU predicts the fragments, and the shapes are compiled in the background. Earlier runs on the same laptop with 0.11.3 (498.7 s CPU) and with the Apple-GPU development build on top of it (389.5 s CoreML, empty cache) are single runs under other conditions: the CoreML difference fits these changes but cannot be attributed to them entirely, and the CPU path is unchanged, so its difference reflects conditions. The generated library content is identical to 0.11.3's on both paths. The laptop CPU library equals the server's up to small numerical differences and rare fragment-selection changes (38 of 5.8 M precursors); the CoreML library has the CPU's RT, CCS and 1/K0 exactly and differs in fragment intensities and rare fragment selections, at the level of CUDA with TF32 off (Astral library: 8 of 3.49 M fragment sets, minimum cosine 0.987).

Fine-tuning. The Astral library of Appendix A (3,488,457 precursors): generation, then tuning the RT model on run 01's 42,785 identifications and re-predicting the whole library.

Table 2: Astral library, generation and fine-tuning (wall time; peak RAM on the host)

configuration	generate (s)	tune RT + re-predict (s)	of which training (s)
server CPU, 96 threads (dax, one socket)	121.7 (8.8 GiB)	181.1 (7.0 GiB)	146 (16 threads)
server, 2 × H100 (data), 16 CPU threads	n/m [3]	51.9 (6.3 GiB)	GPU
laptop CPU, 16 threads, 0.11.4	235.8 (5.4 GiB)	638.8 (5.8 GiB)	578 (8 threads)
laptop Apple GPU, CoreML (warm cache) / Metal, 0.11.4	157.5 (6.4 GiB)	70.8 (6.9 GiB)	40 (GPU)

[3] The server-GPU generation of this library could not be timed cleanly: another user's job occupied one of the GPUs. Writing the DIA-NN TSV took 16 s on the server.

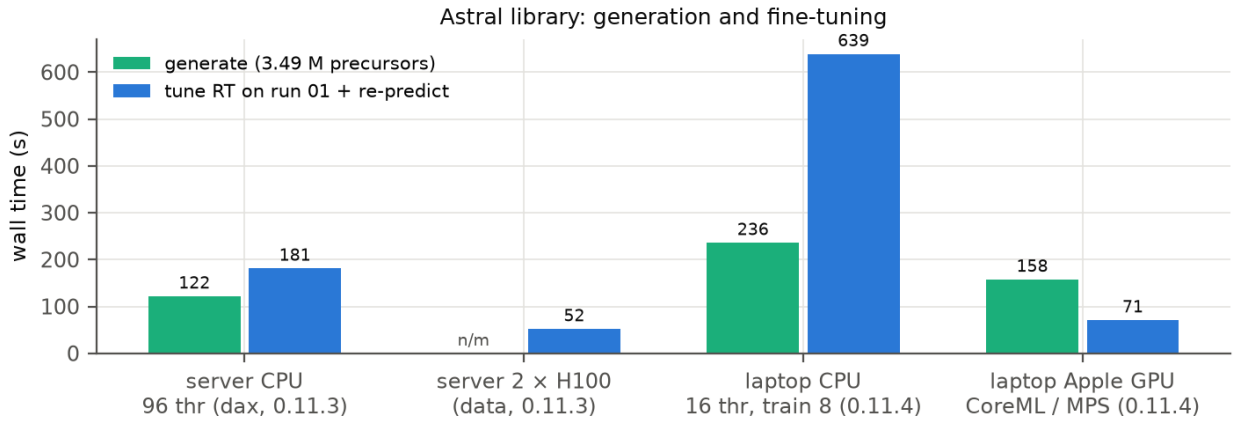


Figure 2: Generation and RT fine-tuning of the 3.49 M-precursor Astral library (servers: DIALibGen 0.11.3; laptop: 0.11.4), wall time per configuration. Tuning starts from run 01's 42,785 library-free identifications and includes training and the whole-library re-prediction, not the DIA-NN search. Servers: 2 × EPYC 9654 (dax) and the same with 2 × H100 (data); laptop: Apple M4 Max. "n/m": not measured.

All four tunes reach the same held-out accuracy: the tune's selection metric (standard deviation of the RT residuals after a straight-line calibration) on the 8,442 held-out test units is 0.393–0.395 min in every case, against 0.607 min for the pretrained model. That metric is the tune's internal score, not the median |ΔRT| of section 4. On the laptop CPU, training is the slow part (578 of 639 s on 8 threads, the macOS default since 0.11.4; 0.11.3 trained on 16 threads in 768 s, with 2,299 s of system time against 501 s now), and training on the Apple GPU, the default on Apple silicon, brings it to 40 s. On this Mac, 8 and 16 training threads gave

identical tuned weights (the documentation states that they may differ). Training is not bit-identical across platforms: the laptop CPU's tuned RTs differ from the server's by a median of 0.022 min (maximum 0.53), the GPU-trained ones from the laptop CPU's by 0.016 min (maximum 0.55). Part of the 0.11.3-to-0.11.4 differences on the laptop may be conditions: some 0.11.3 runs started at a higher load.

DIA-NN 1.7.12. The DIA-NN 1.7.12 available on our cluster is the open-source Linux build, whose deep-learning predictor is not included. Given only a FASTA it writes a library with 0 precursors, so it has no library generation to time. It is used in section 5 for library-free searches and for searches with DIALibGen's libraries.

4. Prediction quality on Astral run 01

Two library sets. This report uses two sets of DIALibGen libraries, generated with the same settings. **Set A** (this section and the DIA-NN 2.0 table of section 5): DIALibGen 0.11.1, tuned on identifications from a search with DIALibGen's own library, so that DIA-NN 2.0 and DIALibGen could be fine-tuned on identical training peptides; it compares the predictors on the same training pool, but that pool came from a search with DIALibGen's library, so a selection bias is possible (see "Training data matter" below). **Set B** (the DIA-NN 1.7.12 table of section 5 and the commands of Appendix A): DIALibGen 0.11.3, tuned on DIA-NN 1.7.12's library-free identifications; it is the recipe a user would run. The two were made at different times for different questions, which is why the 1.7.12 and 2.0 searches do not share libraries.

Libraries compared (set A). Same FASTA and digest in both tools (Appendix A settings; DIA-NN with the equivalent flags):

- DIALibGen, not tuned, and DIALibGen fine-tuned on run 01 (RT model only, whole library re-predicted);
- DIA-NN 2.0's own predicted library (`--predictor --gen-spec-lib`), and DIA-NN 2.0 fine-tuned on the same run-01 training peptides (`--tune-lib … --tune-rt --tune-lr 0.0001`, then re-predicted with the tuned model).

The DIALibGen libraries here were made with 0.11.1, whose untuned library is identical to 0.11.3's. Both tunes used the same training peptides. A DIA-NN 1.7.12 search of run 01 with DIALibGen's untuned library gave 84,076 precursors at 1 % FDR, 77,029 modified sequences. Split by protein group as in section 1, 15,297 sequences form the held-out test set, 7,732 the validation set and 54,000 (58,782 precursors) the training pool, which both tools were fitted on. DIALibGen used its default tune settings with `-machine:threads 64` and chose its best epoch on the validation set; DIA-NN's tuner has no validation set and a fixed schedule, so it ran at learning rate 1e-4 on the training pool alone and was given neither the validation nor the test sequences. Each tool was tuned once.

Retention time

Reference RTs. Run 01 was searched with DIA-NN 2.0 twice, once with each untuned library. A precursor is a reference when it is identified at 1 % FDR in both searches and both apexes agree within 0.1 min (99.7 % of the shared IDs do); its observed RT is the mean apex. Only precursors present in all libraries are scored (80,886). Which precursors enter the reference thus depends on the two untuned libraries; the apex itself is measured from the data, and requiring both searches to agree keeps precursors that neither library misses.

Held out. Proteins are split into cohorts by a hash of their protein group. About one in five belongs to the test cohort; neither tool's tuning saw these proteins, and the scores below use only their 16,068 precursors.

Calibration. Library RT is on different scales (DIALibGen 0–1, DIA-NN iRT, tuned libraries run-01 minutes), and a search engine maps it to the run. Two maps are applied here, both fitted on calibration precursors and scored on others: a straight line (slope and intercept, the classic iRT-style calibration), and a monotone LOWESS fit (locally weighted regression, span 5 %), which also absorbs a non-linear gradient. On run 01 the maps are fitted on the non-held-out precursors and scored on the held-out ones; on runs 02 and 03 they are fitted on one half of the reference precursors (split by peptide sequence) and scored on the other half. That

half is held out from the calibration map but not from the run-01 tuning, since most of its proteins were in the tuning pool. Restricted to the proteins that were also held out from tuning (16,331 observations), the pooled tuned-as-stored median is 0.095 min instead of 0.083. ΔRT = library RT after the map − observed RT; the "95th percentile" of |ΔRT| is the value below which 95 % of the precursors lie.

Stages. ΔRT is reported at each stage of the workflow: (1) the predicted library with the linear map; (2) the predicted library with the LOWESS map; (3) the tuned library as stored, in run-01 minutes, without any map, i.e. its agreement with the run before a search engine's own RT alignment; (4) the tuned library with the LOWESS map; (5) the refined library, in which the precursors identified in run 01 carry run 01's observed RT, scored on runs 02 and 03 against their observed RT (on run 01 itself it is in-sample); (6) `refine –tune –no_filter`, the identified precursors observed and the rest tuned, scored on runs 02 and 03. The refined library is `–mode refine` of the DIA-NN 1.7.12 run-01 report that also supplied the training peptides (84,076 precursors at 1 % FDR, observed apex RT; `–q_global 1 –q_protein 1`); stage 6 was assembled from its values and the tuned library's, not exported as a file. Stages 1–4 are also given for DIA-NN 2.0's predicted and tuned libraries.

Table 3: ΔRT by stage, the main rows (minutes). Runs 02 and 03 are technical replicates and are pooled here. Appendix B has both tools at every stage, the 68th percentile and the ±1 min share, and the two runs separately.

stage: library RT and mapping	n	median |ΔRT| (min)	95th per-centile (min)	within ±0.5 min
Run 01: 16,068 precursors of proteins held out from tuning				
DIALibGen, stage 1: predicted, linear fit	16,068	0.334	1.203	68.4 %
DIALibGen, stage 2: predicted, LOWESS	16,068	0.243	0.856	81.1 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes)	16,068	0.094	0.351	98.3 %
DIALibGen, stage 4: tuned, LOWESS	16,068	0.093	0.349	98.4 %
DIA-NN 2.0, stage 2: predicted, LOWESS	16,068	0.290	0.919	74.4 %
DIA-NN 2.0, stage 4: tuned, LOWESS	16,068	0.142	0.484	95.5 %
Runs 02 and 03 pooled, evaluation halves (held out from the maps, not from the tuning; n = observations)				
DIALibGen, stage 3: tuned, as stored (run-01 minutes)	82,223	0.083	0.345	98.1 %
DIALibGen, stage 4: tuned, LOWESS	82,223	0.081	0.334	98.4 %
stage 5: refined, observed run-01 RT, as stored (identified in run 01)	66,635	0.006	0.079	99.6 %
stage 4: tuned, LOWESS, on the same precursors	66,635	0.077	0.314	98.7 %
stage 6: refine -tune -no_filter, precursors not identified in run 01 (tuned, as stored)	15,588	0.101	0.434	96.5 %
stage 6: refine -tune -no_filter, all precursors, as stored	82,223	0.016	0.210	99.0 %

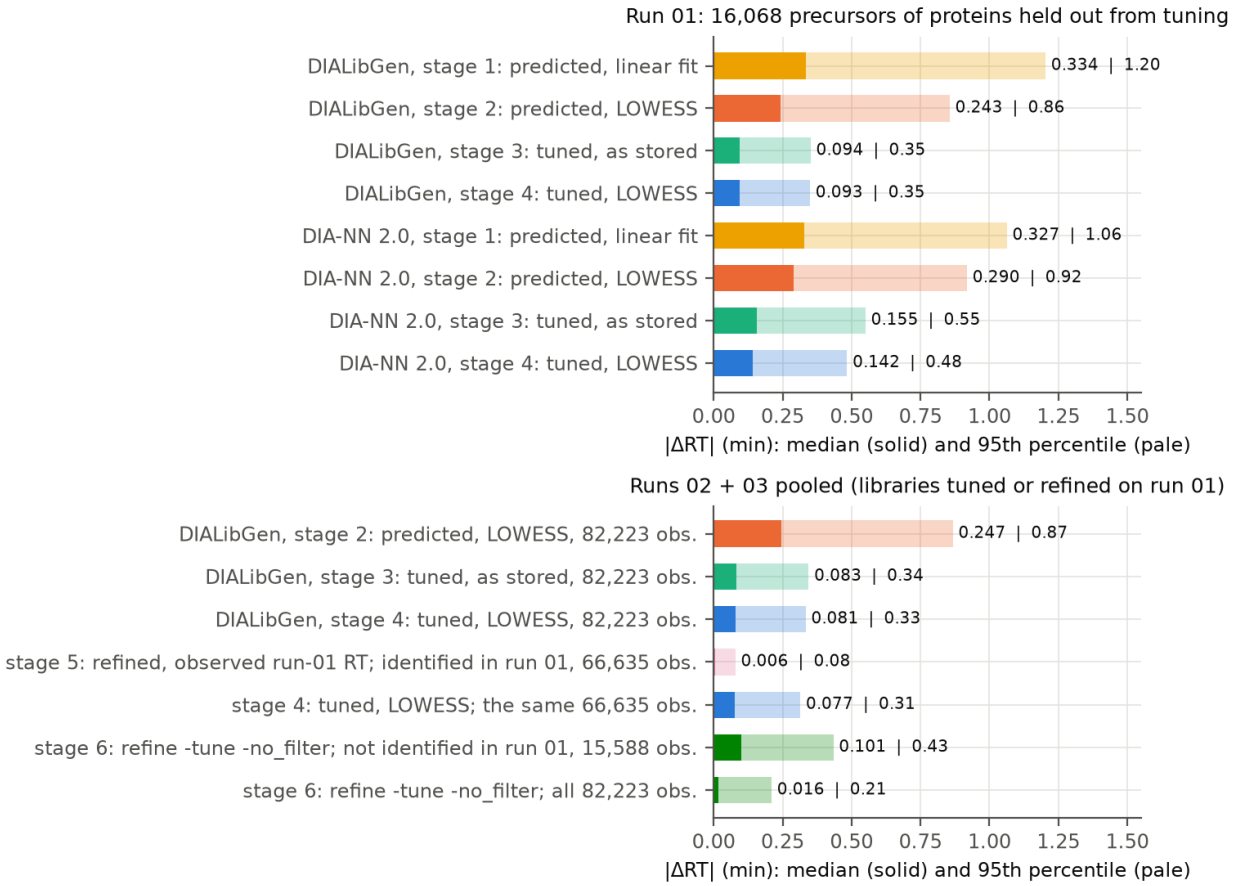


Figure 3: Absolute RT error by stage, in minutes: solid bars are medians, pale bars 95th percentiles of |ΔRT|. Linear and LOWESS maps are fitted on separate calibration precursors; "as stored" uses no map. Top: run 01, 16,068 precursors of proteins excluded from tuning, both tools. Bottom: runs 02 and 03 pooled, DIALibGen; the evaluation observations are held out from the maps but include tuning proteins; full-set rows use 82,223 observations, the identified subset 66,635 and the not-identified subset 15,588. Stage 5 is run 01's observed RT carried over to runs 02/03.

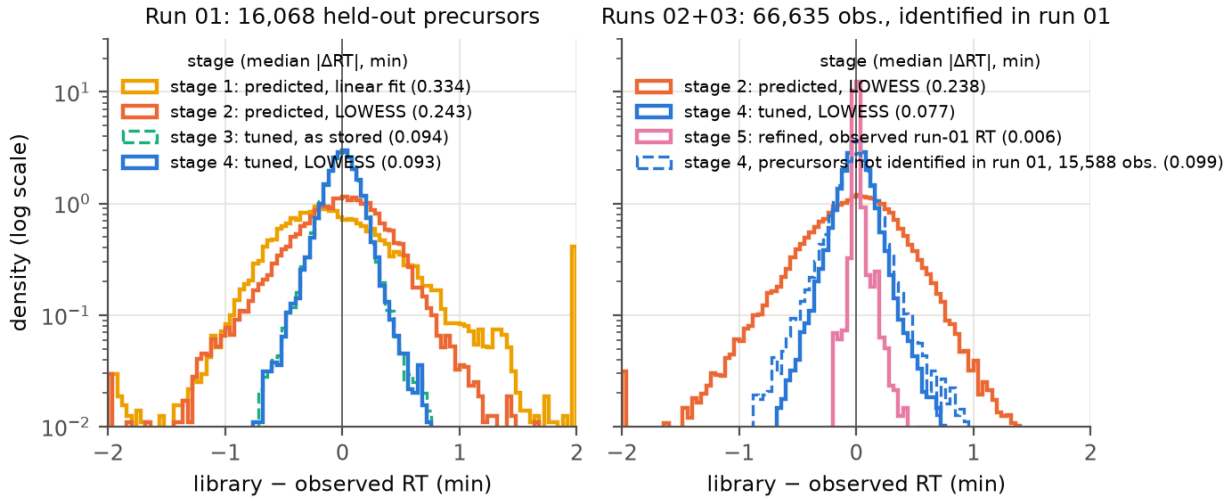


Figure 4: Distributions of DIALibGen's RT residuals (library RT minus observed RT, minutes) by stage, histograms normalised separately on a logarithmic density axis; values beyond ±2 min are collected in the edge bins. Left: run 01, 16,068 precursors of proteins excluded from tuning, stages 1–4. Right: runs 02 and 03 pooled, 66,635 observations of precursors identified in run 01, stages 2, 4 and 5; dashed: stage 4 for the 15,588 observations of precursors not identified in run 01.

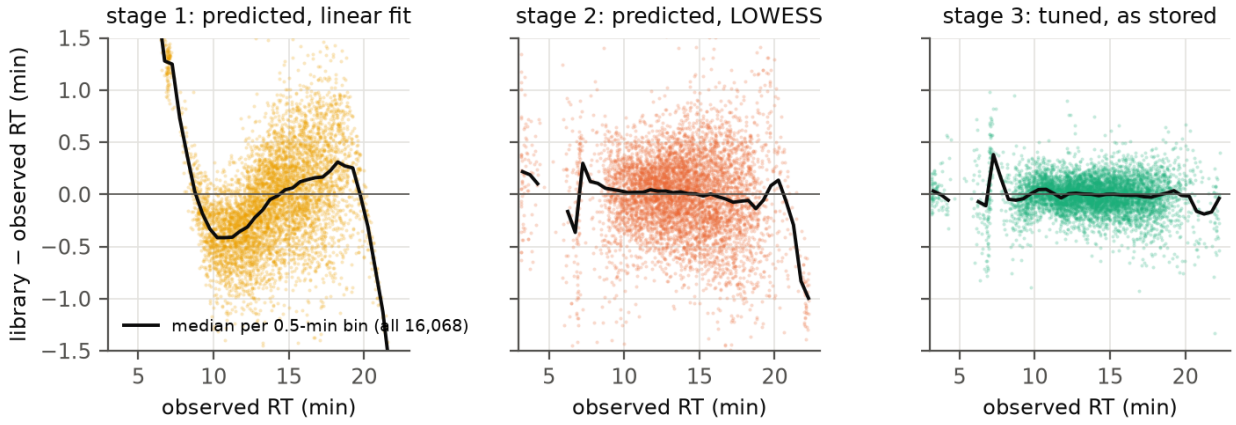


Figure 5: DIALibGen's RT residual against observed RT on run 01, for precursors of proteins excluded from tuning: 6,000 of the 16,068 are shown as points, the black line is the median in each 0.5-min bin over all 16,068. Panels: linear calibration (stage 1), LOWESS calibration (stage 2), tuned library without calibration (stage 3). The linear fit leaves a curved residual, the LOWESS map removes most of it, and the tuned library is nearly flat without any map.

Reading the stages:

- **1 → 2, the predicted RT scale is not linear in run time.** A straight-line calibration of the predicted library leaves a curved residual (ΔRT-vs-RT figure, left panel) and a median error of 0.334 min; the monotone map brings it to 0.243 min. DIA-NN's iRT scale is closer to linear in run time (0.327 → 0.290 min), DIALibGen's raw output less so, but after the map DIALibGen's prediction is the more accurate.
- **2 → 3, tuning does most of the work; 3 → 4, a map changes little.** The tuned library used as stored, without any map, is already at 0.094 min on run 01 and 0.083 min on runs 02/03; the map changes the median by 0.001–0.002 min (on run 03 it also trims the 95th percentile from 0.357 to 0.337 min and removes a +0.011 min offset), and some residual structure remains at the ends of the gradient (ΔRT-vs-RT figure, right panel). DIA-NN's tuned library still gains from a map (0.155 → 0.142 min) and has a

−0.02 min offset as stored.

- **5, observed values on a sibling run.** For the precursors identified in run 01, run 01's observed RT predicts runs 02/03 within a median of 0.006 min and 95 % within 0.08 min: run-to-run reproducibility of this LC setup, a median error about 12-fold lower than the tuned prediction on the same precursors (0.077 min). These precursors are, however, only those found in run 01 (81 % of the pooled reference observations here); a refined library cannot identify the rest.
- **6, both at once.** `refine –tune –no_filter` keeps the whole library: the identified 81 % behave as stage 5, the remaining 19 % as the tuned library (0.101 min as stored; these are the harder precursors, slightly above the 0.083 min of all, and about a tenth of them are peptides seen in run 01 at another charge state). The combined library has a median error of 0.016 min on the pooled evaluation set of a sibling run.
- **Recommendation.** Tune for a complete library whose RT can be used as stored; refine when a run-specific empirical library of the identified precursors is wanted; `refine –tune –no_filter` gives observed accuracy where the run supports it and tuned predictions everywhere else. On this LC setup a tuned library's stored RT already agrees with the run before the search engine's own RT alignment; a predicted library depends on that alignment handling the non-linearity (stage 1 is the straight-line baseline). Identification yields of refined and hybrid libraries were not benchmarked; section 5 covers predicted and tuned libraries only.
- **Training data matter.** The shared training peptides came from a search with DIALibGen's library, which may suit DIALibGen's tuner. DIA-NN tuned on the identifications of its own DIA-NN 2.0 run-01 search (70,645 instead of 58,782 training precursors, validation proteins included) reached 0.099 min (LOWESS-calibrated), close to DIALibGen's 0.093 min. Training source, size and validation membership changed together, so the gain cannot be attributed to the source alone; tuning variability was not measured (one tune per tool).
- Set B (0.11.3, the DIA-NN 1.7.12 table of section 5) used DIA-NN 1.7.12's library-free identifications (42,785 precursors) both for tuning and as the RT reference. Median error on run 01's held-out proteins: 0.225 min untuned, 0.101 min tuned. Different training data and reference, so not directly comparable with the table above.

Fragment intensities

Observed intensities. DIA-NN 1.7.12 searched run 01 without a library. No predicted intensities enter this search, so it does not take the observed ions or their intensities from either library under test; it is not proven free of every bias (which precursors it finds and how it quantifies them are its own). It reported, with `--report-lib-info`, each quantified fragment's identity and raw quantity. For each of its 42,785 precursors at 1 % FDR it reports up to six y ions (median 6). This is the library-free search of section 5, repeated with the fragment export; it found the same 42,785 precursors.

Comparison. For each precursor, the observed fragment quantities were compared with each library's predicted intensities for the same ions (type, number, charge). Every observed ion matched by a library was checked against that library's m/z: all 411,854 ion matches (an ion matched by both libraries counts twice) agreed within 0.02 Th. Scores: cosine similarity, and the normalised spectral angle $1 - 2 \cdot \arccos(\cos)/\pi$, which spreads out values near 1. The main analysis uses, per precursor, only the observed ions that both libraries store (each stores up to 12 of its own choice); requiring at least three such ions leaves 40,419 precursors, with a median of 5 shared ions. Fragment intensities are not changed by tuning, so this applies to tuned and untuned libraries alike.

Table 4: Predicted vs observed y-ion intensities, run 01

library	precursors	median cosine (interquartile range)	cosine ≥ 0.9	median spectral angle (IQR)	higher cosine than the other library
DIALibGen	40,419	**0.986** (0.955–0.996)	88.4 %	**0.892** (0.807–0.941)	69.1 % of precursors
DIA-NN 2.0 predicted	40,419	0.977 (0.942–0.991)	86.3 %	0.864 (0.782–0.916)	30.9 %

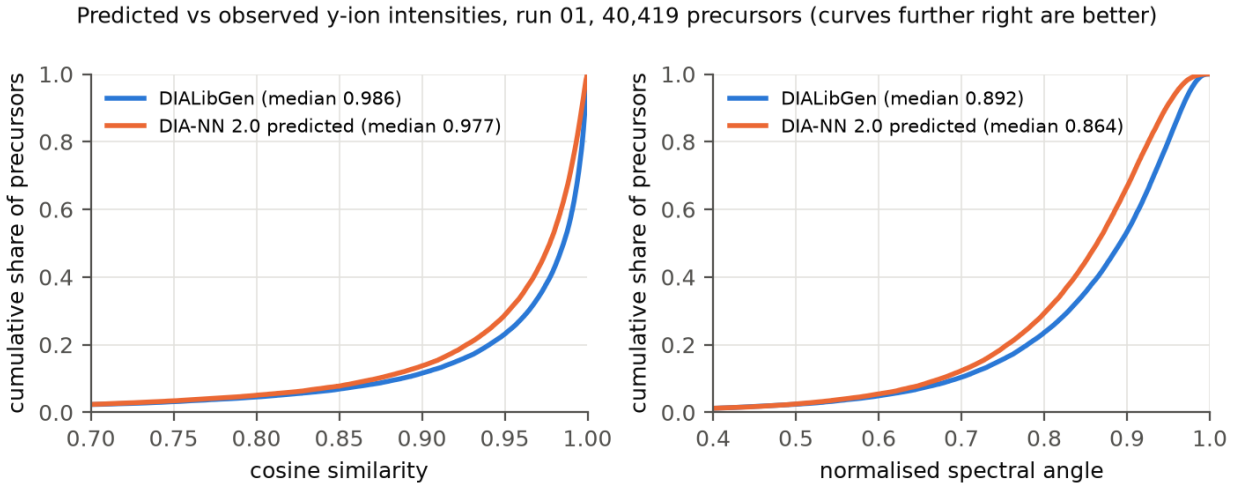


Figure 6: Agreement of predicted with observed y-ion intensities for 40,419 precursors of Astral run 01, as cumulative distributions: cosine similarity (left) and normalised spectral angle (right); a curve further right is better. Observed fragment quantities come from a library-free DIA-NN 1.7.12 search; each precursor is scored on at least three observed ions stored by both libraries (median five).

- DIALibGen's predicted intensities agree better with the observed ones, by a small margin that holds for most precursors (median per-precursor difference +0.004 cosine).
- Two variants give the same result: scoring each library on all observed ions it stores, at least three (40,445 precursors have that many in both; 0.986 vs 0.977), and scoring on every observed ion with a missing library ion counted as intensity 0 (42,344 precursors; 0.984 vs 0.977, spectral angle 0.887 vs 0.862).
- By charge: 2+ (37,896 precursors) 0.986 vs 0.978, 3+ (2,177) 0.979 vs 0.973; 1+ and 4+ are too few to compare.
- **Limits of this check.** Only y ions, at most six per precursor, were observed; b-ion predictions are not tested. The precursors are those a library-free search finds, mostly doubly charged and abundant. DIALibGen was run with the Astral setting (NCE 25); DIA-NN's predictor has no instrument setting.

5. Identifications

All searches: each run on its own, no match-between-runs. Precursors: distinct precursor ids at `Q.Value` ≤ 0.01; protein groups: distinct `Protein.Group` values with `PG.Q.Value` ≤ 0.01 among those precursors. The counts are whole-run identifications; the proteins used for tuning are included.

Table 5: DIA-NN 1.7.12 with library set B (DIALibGen 0.11.3) and library-free

library	run 01	run 02	run 03	protein groups (01 / 02 / 03)
none (DIA-NN 1.7.12 library-free)	42,785	42,586	41,695	5,577 / 5,717 / 5,493
DIALibGen, not tuned	84,161	83,239	83,978	7,538 / 7,607 / 7,613
DIALibGen, fine-tuned on run 01	90,253	92,090	87,838	7,898 / 8,088 / 7,877

The 1.7.12 baseline is library-free, not a predicted DIA-NN library: this build cannot predict (section 3), so its search has no predicted intensities and the same library RT for every precursor. The comparison shows what a predicted library adds to DIA-NN 1.7.12, not how DIALibGen compares with DIA-NN's predictor. In each run, 97.5–97.7 % of the library-free search's precursors are also found with the tuned library, which finds about twice as many in total. Run 01 supplied the training data of the tuned library; runs 02 and 03 are technical replicates. (The untuned-library search that supplied set A's training data found 84,076 precursors in run 01 instead of 84,161: its TSV had been converted with an earlier convention that rescaled RT to 0–100, and DIA-NN's RT map is slightly sensitive to the scale; 0.1 % of the identifications differ.)

Table 6: DIA-NN 2.0 with each predicted library (library set A, section 4)

library	run 01	run 02	run 03
DIA-NN 2.0, predicted	88,130	89,704	88,432
DIALibGen, not tuned	89,232	90,362	89,318
DIA-NN 2.0, fine-tuned on run 01	90,843	92,283	90,616
DIALibGen, fine-tuned on run 01	96,156	96,040	95,552

Untuned, the two predicted libraries give similar numbers (DIALibGen +0.7 to +1.3 %); after fine-tuning, DIALibGen's gives 4–6 % more precursors. Run 01 supplied the training data; runs 02 and 03 are technical replicates. These are DIA-NN's own 1 % FDR estimates; they were not checked with entrapment, and a library that narrows the RT window can change how well that estimate holds. They are precursor counts: protein-group gains and quantitative accuracy were not assessed for DIA-NN 2.0.

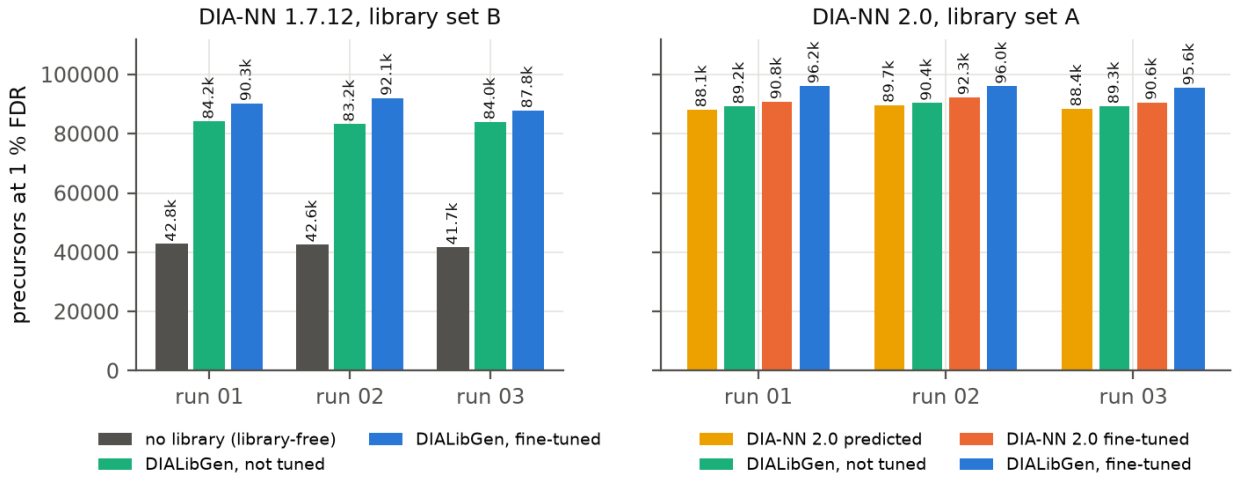


Figure 7: Whole-run precursor identifications passing DIA-NN's estimated 1 % FDR, per run, without match-between-runs or entrapment validation; all tuned libraries were trained on run 01, runs 02 and 03 are technical replicates. Left: DIA-NN 1.7.12 with no library and with library set B (DIALibGen 0.11.3, tuned on library-free identifications). Right: DIA-NN 2.0 with the four predicted libraries of set A (both predictors tuned on the shared training pool).

Entrapment check of the 1 % FDR

Design. Five of the seven search configurations of this section (DIA-NN 2.0's own tuned library and DIA-NN 1.7.12 with the untuned library were left out) were repeated on ProteoBench's entrapment database for these runs: a pre-digested FASTA of 1,420,868 human target peptides (contaminants included) and, for each, one entrapment twin with the same amino-acid composition in shuffled order (ratio r = 1, paired 1:1; the twins cannot be in the sample). No in-silico digestion was applied anywhere; otherwise the settings of the searches above were kept, except that the reports were written down to $q \leq 0.05$ so that the matched-FDP cut-off can lie above q = 0.01 (the counts at $q \leq 0.01$ equal DIA-NN's own log counts in every search). Libraries: DIA-NN 2.0's predicted library of the entrapment FASTA (7,008,740 precursors); DIALibGen 0.11.1 untuned and tuned (set-A recipe: tuned on run 01's identifications from a DIA-NN 1.7.12 search of the untuned library; 7,007,990 precursors); DIA-NN 1.7.12 library-free; and DIALibGen 0.11.3 tuned on the library-free identifications of run 01 (set-B recipe). Each precursor at the cut-off is a target or an entrapment hit by its stripped sequence. With T targets and E entrapment hits, the combined estimate is $FDP = E(1 + 1/r) / (T + E) = 2E / N$, with a Wilson 95 % interval on E/N scaled by 2 (an approximate interval for the estimate, not for the true FDP). The paired estimate (Wen et al. 2025) additionally counts an entrapment precursor whose twin precursor (same pair, charge and number of oxidised methionines) is absent, or ranks better than its twin, as evidence of a false target; both estimates are upper bounds under the entrapment assumptions, the lower bound is E / N. "Targets at matched 1 % FDP" ranks all reported precursors by q-value (ties by PEP or CScore) and takes the largest prefix whose combined FDP is at most 1 %: a retrospective calibration on the entrapment labels, used here to compare the libraries descriptively at the same estimated FDP, not a prospective FDR control. Pooled rows count precursor–run observations. Arms with an FDP above 1.5 % are marked **!**.

Table 7: Entrapment FDP at nominal $q \leq 0.01$, precursor level, per run and pooled

arm	run	targets, $q \leq 0.01$	entrap. hits	combined FDP (95 % CI)	paired FDP	FDP / 1 %	targets, matched 1 % FDP
DIA-NN 2.0, its own predicted library	01	81,052	353	0.87 % (0.78–0.96)	0.83 %	0.87	82,953
	02	79,172	311	0.78 % (0.70–0.87)	0.74 %	0.78	81,707
	03	79,909	313	0.78 % (0.70–0.87)	0.73 %	0.78	82,421
	01–03 pooled	240,133	977	0.81 % (0.76–0.86)	0.77 %	0.81	246,842
DIA-NN 2.0, DIALibGen untuned	01	81,967	351	0.85 % (0.77–0.95)	0.80 %	0.85	83,536
	02	83,406	296	0.71 % (0.63–0.79)	0.68 %	0.71	86,382
	03	81,429	338	0.83 % (0.74–0.92)	0.77 %	0.83	83,401
	01–03 pooled	246,802	985	0.80 % (0.75–0.85)	0.75 %	0.80	253,227
DIA-NN 2.0, DIALibGen fine-tuned on run 01	01	86,243	388	0.90 % (0.81–0.99)	0.85 %	0.90	87,315
	02	86,845	325	0.75 % (0.67–0.83)	0.71 %	0.75	89,041
	03	85,825	345	0.80 % (0.72–0.89)	0.76 %	0.80	87,938
	01–03 pooled	258,913	1,058	0.81 % (0.77–0.86)	0.77 %	0.81	264,335
DIA-NN 1.7.12, library-free	01	33,312	177	1.06 % (0.91–1.22)	0.98 %	1.06	32,773
	02	37,922	209	1.10 % (0.96–1.25)	1.06 %	1.10	37,436
	03	34,359	175	1.01 % (0.87–1.17)	0.97 %	1.01	34,280

arm	run	targets, q ≤ 0.01	entrap. hits	combined FDP (95 % CI)	paired FDP	FDP / 1 %	targets, matched 1 % FDP
	01–03 pooled	105,593	561	1.06 % (0.97–1.15)	1.01 %	1.06	104,480
DIA-NN 1.7.12, DIALibGen fine-tuned on run 01	01	78,504	2,202	5.46 % (5.24–5.69) !	5.30 %	5.46	68,809
	02	78,837	1,978	4.90 % (4.69–5.11) !	4.73 %	4.90	70,722
	03	76,799	2,543	6.41 % (6.17–6.66) !	6.30 %	6.41	67,155
	01–03 pooled	234,140	6,723	5.58 % (5.45–5.72) !	5.44 %	5.58	206,003

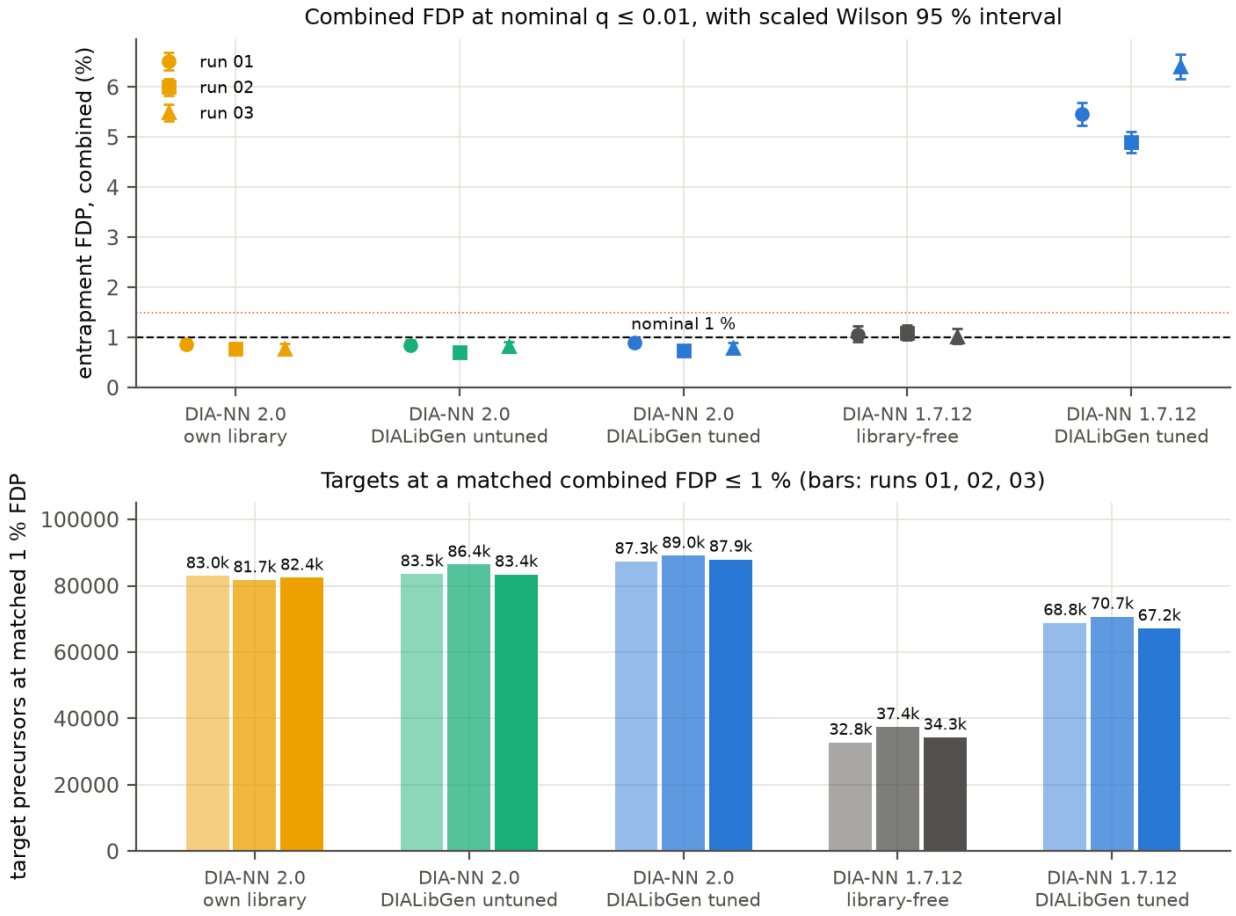


Figure 8: Entrapment results per arm and run. Top: combined FDP at nominal q ≤ 0.01 with its scaled Wilson 95 % interval (symbols: runs 01, 02, 03); dashed line nominal 1 %, dotted 1.5 %. Bottom: target precursors at a matched combined FDP of at most 1 % (bars left to right: runs 01, 02, 03).

- **DIA-NN 2.0 stays within its 1 % with all three libraries tested.** Combined FDP 0.71–0.90 % for its own predicted library and for DIALibGen's untuned and tuned libraries alike: in DIA-NN 2.0, DIALibGen's libraries are consistent with 1 % control to the same degree as DIA-NN's own (similar estimates, not a proof of equivalence).
- **At a matched 1 % FDP the ranking of section 5 holds:** pooled over the runs, DIALibGen tuned

identifies 264,335 targets, DIALibGen untuned 253,227 and DIA-NN's library 246,842 (+7.1 % and +2.6 % over DIA-NN's library; precursor–run observations, the matched cut chosen on the pooled data). This is the fairer headline than the counts at nominal q.

- **DIA-NN 1.7.12 understates its error rate with a predicted library.** Library-free it is approximately consistent with nominal control (combined 1.01–1.10 %, intervals including 1 %), but with DIALibGen's tuned library its nominal 1 % corresponds to an estimated 4.9–6.4 % (even the lower bound is 2.4–3.2 %). This reproduces the 5.2 % seen earlier for the untuned 0.11.1 library in this setup (2,047 entrapment hits among 78,405 at $q \leq 0.01$ on run 01). The FDP of the set-B searches of the human proteome (the table above) was not measured; this result is a strong indication that their nominal counts are too high. At a matched 1 % FDP the tuned library still yields 206,003 targets against 104,480 library-free (2.0 ×), reached at $q \approx 0.003$.
- Peptide-level estimates (distinct modified sequences) give the same picture (pooled combined FDP 0.86–0.88 % for the DIA-NN 2.0 arms, 1.09 % library-free, 6.0 % for 1.7.12 with the tuned library). Protein-group estimates are not meaningful here: every FASTA entry is a single peptide.
- Caveats: one dataset and one entrapment design (shuffled twins); the entrapment FASTA has no protein context, so the searches differ from section 5 in search space (two sequences per target peptide) and the identification counts are not comparable with those tables; the tuned libraries were trained on run 01, so run 01 is in-sample for them (each run is searched and RT-aligned on its own); the matched cut uses the entrapment labels of the same data.

6. Limitations

- **Speed at moderate thread counts.** At 16 threads DIALibGen 0.11.3 (and 0.11.4 and 0.11.5, which keep its CPU path) is slower than 0.11.2 and than DIA-NN 2.0 (section 3); it was faster only at 96 threads. DIA-NN on a GPU was not measured.
- **GPU use**: since 0.12.1 the Linux packages predict on NVIDIA GPUs with the user's CUDA 12 runtime and cuDNN 9 (Appendix A, "Threads and GPUs"); the linux-arm64 GPU route has not been run on NVIDIA hardware; fine-tuning on NVIDIA GPUs needs a source build with CUDA LibTorch. The Apple GPU is supported by the macOS release, for the fragment model and training only.
- **OpenSWATH PQP export** holds the whole library in memory: a whole human proteome with charges 1–4 (about 70 M transitions) needs about 60 GB RAM.
- **Identifications from DIALibGen's own search of the run (`--run`) are not yet validated**; use a DIA-NN report (`--ids`).
- **Scope of the comparisons.** Accuracy and identifications come from one Astral dataset (three technical replicates, one 15-min gradient). Ion-mobility (CCS) prediction and tuning were not evaluated. Each timing is a single run (three for one cell), and DIA-NN and DIALibGen 0.11.3 were timed on two identical nodes.
- **OpenSWATH output** was not used in a search here; a calibrated, scored OpenSWATH search with a DIALibGen library has not been benchmarked.
- Fragment-intensity agreement was measured on y ions only. DIALibGen predicts b/y ions without neutral losses and does not fine-tune the fragment model.

7. Availability

- **Source and issues:** https://github.com/okohlbacher/DIALibGen
- **Release 0.12.1:** https://github.com/okohlbacher/DIALibGen/releases/tag/v0.12.1 (all releases: https://github.com/okohlbacher/DIALibGen/releases)
- **Homebrew tap:** https://github.com/okohlbacher/homebrew-dialibrarygenerator
- **Licence:** BSD-3-Clause. Bundled third-party components keep their own licences (listed in the repository's third-party notices).
- **How to cite:** cite the DIALibGen repository and version, and the methods it builds on: AlphaPeptDeep (Zeng et al., *Nature Communications* 13, 7238, 2022) and OpenMS (Röst et al., *Nature Methods* 13,

741–748, 2016).

- **Contact:** Oliver Kohlbacher, University of Tübingen; questions and bug reports through the GitHub issue tracker.
- **This report's analysis scripts and data** (fragment-intensity comparison, the RT comparison, figures, per-precursor values) are in the `scripts/` and `data/` folders of the report bundle; there is no public archive yet, so request them from the author.
- **Software used here:** DIALibGen 0.11.1 (release build), 0.11.2 (server runtime rows), 0.11.3 (built from source against OpenMS 3.5.0, ONNX Runtime 1.30.0 and LibTorch 2.10; GPU build with CUDA 12.8 and cuDNN 9), 0.11.4 (macOS release from the Homebrew cask), 0.11.5 (GPU build as for 0.11.3) and 0.12.1 (Linux x64 release package with ONNX Runtime's CUDA build; cuDNN 9 from NVIDIA), the AlphaPeptDeep RT, MS2 and CCS models bundled with the 0.11.1 release (ONNX files `peptdeep_{rt,ms2,ccs}_dynamic.onnx`; their checksums are embedded in every Parquet library), DIA-NN 2.0 (Academia, 28 January 2025) and DIA-NN 1.7.12 (open-source Linux build).

Appendix A. Installation and usage

Installation

- **Release archives and installers:** https://github.com/okohlbacher/DIALibGen/releases. The CLI archive (`DIALibGen-<platform>.tar.gz` / `.zip`) holds `bin/`, `lib/` and `share/`; keep them together. Desktop app: `.dmg` (macOS), `.AppImage`/`.deb` (Linux), `.msi`/`.exe` (Windows).
- **Homebrew** (macOS 14 or later), from the tap `okohlbacher/homebrew-dialibrarygenerator`: `brew install --cask okohlbacher/dialibrarygenerator/dialibgen-cli` (CLI) and `brew install --cask okohlbacher/dialibrarygenerator/dialibgen` (desktop app).
- **From source:** CMake, a C++23 compiler, OpenMS 3.5, Arrow/Parquet, ONNX Runtime and, for fine-tuning, LibTorch; see `docs/building.md` in the repository. The Linux release packages predict on NVIDIA GPUs since 0.12.1 (see "Threads and GPUs" below; training there needs a source build); the macOS release uses the Apple GPU as it is.

`DIALibGen --helphelp` lists every option; `docs/usage.md` and `docs/parameters.md` explain them.

Commands used for the benchmark libraries

These are the commands of the Astral libraries of section 5 and the fine-tuning times of section 3, written for DIALibGen 0.12.1. The libraries and times were produced with DIALibGen 0.11.3 (release models; 0.11.4 generates the same library content on the CPU) and DIA-NN 1.7.12 (open-source Linux build); 0.11.3 needed every setting of the table below on the command line (`-generation:nce 25 -generation:precursor_mz 380 980 -generation:fragment_mz 150 2000 -generation:precursor_charges 1 2 3 4` and the defaults), and the 0.12.1 command writes the same TSV, byte for byte (checked on the CPU with a 3,000-protein FASTA). The libraries of section 4 were generated with the same settings but tuned on a different set of identifications (section 4). Only the paths are shortened; `-threads 96` is what was timed (0.12.1 uses all physical cores without it).

```
# 1. predict the library from the FASTA (122 s at 96 threads, 0.11.3); astral.yaml
#    supplies NCE 25, the m/z windows and charges 1–4
DIALibGen -in UP000005640_9606.fasta -out lib.parquet -generation:instrument Astral \
  -generation:variable_modifications 'Oxidation (M)' -threads 96

# 2. first-pass DIA-NN search of run 01 (here: DIA-NN 1.7.12, no library)
DIGEST="--cut K*,R* --missed-cleavages 1 --min-pep-len 7 --max-pep-len 30
  --min-pr-charge 1 --max-pr-charge 4 --min-pr-mz 380 --max-pr-mz 980
  --min-fr-mz 150 --max-fr-mz 2000 --unimod4 --var-mods 1 --unimod35
  --met-excision"
mkdir -p run01/temp      # DIA-NN 1.7.12 stops if --temp does not exist
diann --f LFQ_Astral_DIA_15min_50ng_Human_01.mzML \
  --fasta UP000005640_9606.fasta --fasta-search $DIGEST --threads 96 \
```

```
  --verbose 1 --qvalue 0.01 --out run01/report.tsv --temp run01/temp
# DIA-NN 1.7.12 only: report.tsv -> report.parquet with precursor m/z
#   (needs Python with numpy, pandas and pyarrow)
python3 scripts/prep_ids.py run01/report.tsv run01.report.parquet lib.parquet

# 3. fine-tune the RT model on run 01, re-predict all 3.49 M precursors
#    (181 s, 0.11.3; it also wrote -tune_out_models tuned-models -out_report residuals.tsv)
DIALibGen -mode tune -in lib.parquet -ids run01.report.parquet -out tuned.parquet -threads 96

# 4. write the DIA-NN TSV (16 s), then search each run N = 1, 2, 3 with it
DIALibGen -in tuned.parquet -out tuned.tsv -threads 96
mkdir -p runN/temp
diann --f LFQ_Astral_DIA_15min_50ng_Human_0N.mzML --lib tuned.tsv \
  --fasta UP000005640_9606.fasta --reannotate $DIGEST --threads 96 \
  --verbose 1 --qvalue 0.01 --out runN/report.tsv --temp runN/temp
```

- `tune` reads a DIA-NN `report.parquet` of **one** run. DIA-NN 2.x writes one directly (`--out report.parquet`). DIA-NN 1.7.12 writes only `report.tsv` and no precursor m/z, so the small script `prep_ids.py` (in this report's `scripts/`) converts it and computes the m/z.
- `tune` chooses what to tune from the report (`-tune_heads auto`): RT here, because the Astral has no ion mobility; RT and CCS for a diaPASEF report with 1/K0. (0.11 tuned both by default and needed `-tune_heads rt`.)
- The tuned library's RT is in run 01's minutes. DIA-NN fits its own RT map, so it can be used as is.
- For DIA-NN 2.x, `-out tuned.tsv` or `-out_format diann2_parquet` work directly from `tune`; the separate conversion is only needed for an existing Parquet library (a library as `-in`, without `-ids` or `-run`, is converted; `-mode convert` says so explicitly).

Library settings (the flags above; everything else at its default):

setting	value
enzyme, missed cleavages	Trypsin/P (cleaves after K/R, also before P), 1
peptide length	7–30 residues
precursor charge, precursor m/z	1–4, 380–980
fragment m/z, fragment charge	150–2000, 1–2
fixed / variable modifications	Carbamidomethyl (C) / Oxidation (M), at most 1 per peptide
N-terminal Met excision	yes
instrument model, NCE	`Astral` (AlphaPeptDeep's Lumos group), NCE 25 (0.12.1: with the windows and charges, from `astral.yaml`)
fragments per precursor	3–12 most intense b/y ions (relative intensity ≥ 1e-4), no neutral losses
decoys	none (the search engine makes its own)
result	3,488,457 precursors, 40,849,390 fragment transitions; Parquet 460 MB, DIA-NN TSV 7.1 GB

To keep these settings as a JSON recipe, add `-write_config recipe.json` to the generation command above (it writes the effective settings and exits); later runs use `-config recipe.json`. The Parquet output also records the recipe and the checksums of the FASTA and models.

Threads and GPUs

Since 0.12, `-threads` defaults to one thread per physical core (0.11 followed the OpenMS default of 1; an INI's value is respected). CPU output does not depend on the thread count. Fine-tuning on the CPU trains on `-machine:train_threads` threads (0.11: `-machine:threads`, still accepted in 0.12; default 16 on Linux and Windows, 8 on macOS); the documentation states that the tuned weights may depend on that number, so keep it fixed when you need identical models. `-machine:device auto` (the default) picks the prediction route and names it in one `device:` line. On Apple silicon it trains on the GPU (Metal) and, from 50,000 precursors

per CPU thread, predicts the fragments there (CoreML); such a library agrees with the CPU library to float rounding, not byte for byte. `-machine:device cpu` keeps everything on the CPU, `-machine:device coreml` predicts the fragments on the GPU at any size, and `DIALibGen -mode prepare` compiles their GPU shapes ahead of a first run. On Linux the 0.12.1 packages include ONNX Runtime's CUDA build but not CUDA or cuDNN: with an NVIDIA GPU, the CUDA 12 runtime and cuDNN 9 installed, `auto` predicts on the GPU; without cuDNN only the fragment model runs there (with a `WARNING` naming the fix); without a GPU it runs on the CPU. The linux-arm64 GPU route has not been run on NVIDIA hardware. `-machine:gpu_devices 0 1` spreads the work over two GPUs, `-machine:require_gpu` turns any CPU fallback into an error, and, in a build with CUDA-enabled LibTorch, `-machine:device cuda:0` trains on the GPU (the re-prediction follows `-machine:device`; 0.11's `-tune_predict_gpu` is deprecated). TensorFloat-32 arithmetic is off by default (`-machine:tf32 false`), so a GPU library matches the CPU library to float rounding.

Appendix B. Commands and details of the accuracy comparison

The libraries of section 4 (DIALibGen 0.11.1, DIA-NN 2.0; `DIGEST` as in Appendix A with `--var-mod UniMod:35,15.994915,M` in place of `--unimod35`). The scripts, including the cohort split and the ΔRT evaluation (`eval_rt.py`: reference, calibration, statistics), are in `scripts/rt-comparison/`.

```
# DIALibGen (run with 0.11.1, written for 0.12.1): generate as in Appendix A, then tune
# on the run-01 report of a DIA-NN 1.7.12 search with that library (84,076 precursors)
DIALibGen -mode tune -in predicted.parquet -ids run01.report.parquet \
  -tune_out_models tuned-models -out tuned.parquet -threads 64 -machine:train_threads 64
# the refined library of stage 5 (observed run-01 RT of the identified precursors;
# 0.11 needed -q_global 1 -q_protein 1, the 0.12 defaults)
DIALibGen -mode refine -in predicted.parquet -ids run01.report.parquet -out refined.parquet

# DIA-NN 2.0: predicted library (D), and its lossless Parquet copy for reading RT
diann --fasta F --fasta-search --predictor --gen-spec-lib $DIGEST --threads 64 \
  --out-lib D.parquet
diann --lib D.predicted.speclib --gen-spec-lib --out-lib D.dump.parquet

# DIA-NN 2.0 fine-tuned on the same training pool (F): tune.parquet holds the
# 54,000 training sequences (scripts/rt-comparison/prep_F.py), then re-predict
diann --tune-lib tune.parquet --tune-rt --tune-lr 0.0001 --threads 64
diann --fasta F --fasta-search --predictor --gen-spec-lib $DIGEST \
  --tokens tune.dict.txt --rt-model tune.tuned_rt.pt --threads 64 --out-lib F.parquet

# every search (each library x runs 01-03), identical flags
diann --f LFQ_Astral_DIA_15min_50ng_Human_0N.mzML --lib LIB --threads 64 --verbose 1 \
  --qvalue 0.01 --gen-spec-lib --rt-profiling --temp tmp --out report.parquet \
  --out-lib emp.parquet
```

Table 8: ΔRT by stage, run 01 (precursors of proteins held out from tuning) and runs 02 and 03 pooled (evaluation halves, held out from the maps, not from the tuning), both tools at every stage (minutes; 68th and 95th percentiles of |ΔRT|)

stage: library RT and mapping	n	median |ΔRT|	68th pct	95th pct	±0.5 min	±1 min
Run 01: 16,068 precursors of proteins held out from tuning						
DIALibGen, stage 1: predicted, linear fit	16,068	0.334	0.496	1.203	68.4 %	92.5 %
DIALibGen, stage 2: predicted, LOWESS	16,068	0.243	0.368	0.856	81.1 %	97.0 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes)	16,068	0.094	0.145	0.351	98.3 %	99.9 %
DIALibGen, stage 4: tuned, LOWESS	16,068	0.093	0.142	0.349	98.4 %	99.9 %
DIA-NN 2.0, stage 1: predicted, linear fit	16,068	0.327	0.491	1.063	68.8 %	93.9 %

stage: library RT and mapping	n	median |ΔRT|	68th pct	95th pct	±0.5 min	±1 min
DIA-NN 2.0, stage 2: predicted, LOWESS	16,068	0.290	0.434	0.919	74.4 %	96.4 %
DIA-NN 2.0, stage 3: tuned, as stored (run-01 minutes)	16,068	0.155	0.233	0.550	93.4 %	99.4 %
DIA-NN 2.0, stage 4: tuned, LOWESS	16,068	0.142	0.216	0.484	95.5 %	99.6 %
Runs 02 and 03 pooled, evaluation halves (n = observations)						
DIALibGen, stage 1: predicted, linear fit	82,223	0.360	0.526	1.241	65.5 %	91.4 %
DIALibGen, stage 2: predicted, LOWESS	82,223	0.247	0.372	0.867	80.4 %	96.8 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes)	82,223	0.083	0.131	0.345	98.1 %	99.8 %
DIALibGen, stage 4: tuned, LOWESS	82,223	0.081	0.127	0.334	98.4 %	99.9 %
DIA-NN 2.0, stage 1: predicted, linear fit	82,223	0.345	0.516	1.121	66.6 %	92.7 %
DIA-NN 2.0, stage 2: predicted, LOWESS	82,223	0.294	0.440	0.945	73.8 %	96.0 %
DIA-NN 2.0, stage 3: tuned, as stored (run-01 minutes)	82,223	0.155	0.236	0.577	92.8 %	99.1 %
DIA-NN 2.0, stage 4: tuned, LOWESS	82,223	0.143	0.215	0.498	95.1 %	99.5 %
stage 5: refined, observed run-01 RT, as stored (identified in run 01)	66,635	0.006	0.022	0.079	99.6 %	99.9 %
DIALibGen, stage 2: predicted, LOWESS, same precursors	66,635	0.238	0.358	0.823	81.8 %	97.3 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes), same precursors	66,635	0.079	0.125	0.324	98.4 %	99.8 %
DIALibGen, stage 4: tuned, LOWESS, same precursors	66,635	0.077	0.122	0.314	98.7 %	99.9 %
stage 6: not identified in run 01, tuned, as stored	15,588	0.101	0.158	0.434	96.5 %	99.5 %
stage 6: not identified in run 01, tuned, LOWESS	15,588	0.099	0.155	0.417	96.9 %	99.8 %
stage 6: all precursors, as stored	82,223	0.016	0.025	0.210	99.0 %	99.8 %

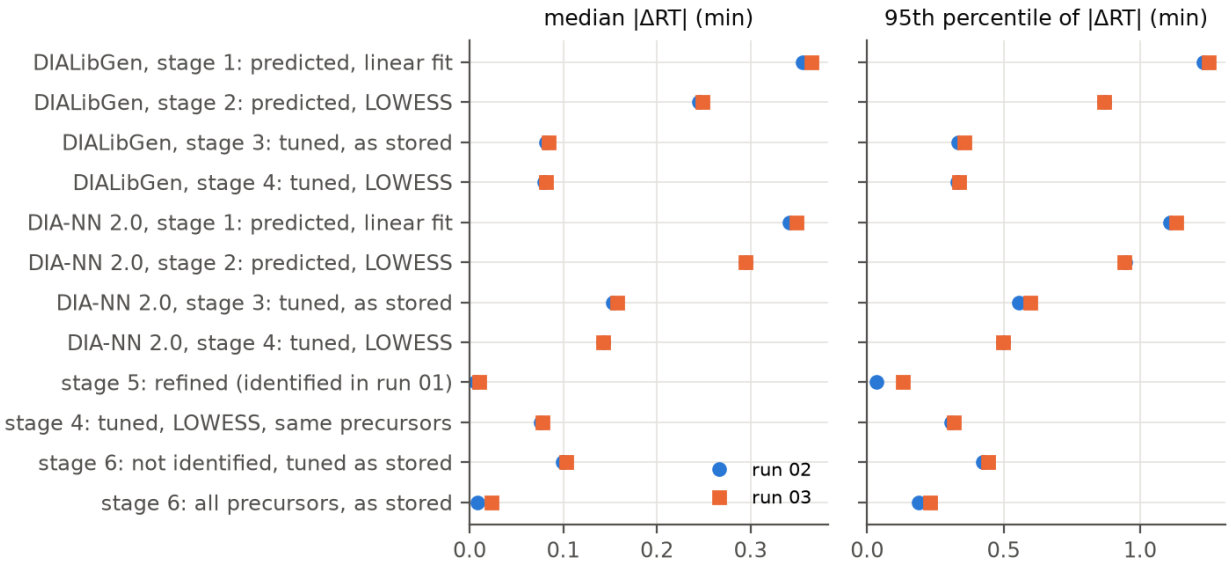


Figure 9: RT error by stage for runs 02 and 03 separately: median |ΔRT| (left) and 95th percentile of |ΔRT| (right), minutes, for the main rows of the table below (the "same precursors" rows of stages 2 and 3 and the stage-6 LOWESS row are omitted). Evaluation observations are held out from the calibration maps; tuning proteins are included.

Table 9: ΔRT by stage, runs 02 and 03 separately (minutes; evaluation halves)

stage: library RT and mapping	n	median |ΔRT|	68th pct	95th pct	±0.5 min	±1 min
Run 02, evaluation half						
DIALibGen, stage 1: predicted, linear fit	41,371	0.356	0.519	1.232	66.1 %	91.7 %
DIALibGen, stage 2: predicted, LOWESS	41,371	0.245	0.371	0.867	80.5 %	96.8 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes)	41,371	0.081	0.128	0.334	98.4 %	99.9 %
DIALibGen, stage 4: tuned, LOWESS	41,371	0.080	0.126	0.329	98.5 %	99.9 %
DIA-NN 2.0, stage 1: predicted, linear fit	41,371	0.342	0.510	1.110	67.2 %	92.9 %
DIA-NN 2.0, stage 2: predicted, LOWESS	41,371	0.294	0.439	0.947	73.9 %	95.9 %
DIA-NN 2.0, stage 3: tuned, as stored (run-01 minutes)	41,371	0.153	0.231	0.556	93.3 %	99.3 %
DIA-NN 2.0, stage 4: tuned, LOWESS	41,371	0.143	0.215	0.497	95.1 %	99.5 %
stage 5: refined, as stored (identified in run 01)	33,470	0.006	0.019	0.036	100.0 %	100.0 %
DIALibGen, stage 2: predicted, LOWESS, same precursors	33,470	0.236	0.357	0.821	82.0 %	97.4 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes), same precursors	33,470	0.078	0.122	0.314	98.8 %	100.0 %
DIALibGen, stage 4: tuned, LOWESS, same precursors	33,470	0.076	0.121	0.310	98.9 %	100.0 %
stage 6: not identified in run 01, tuned, as stored	7,901	0.099	0.156	0.424	96.8 %	99.8 %
stage 6: not identified in run 01, tuned, LOWESS	7,901	0.098	0.154	0.413	96.9 %	99.8 %
stage 6: all precursors, as stored	41,371	0.008	0.022	0.188	99.3 %	99.9 %
Run 03, evaluation half						
DIALibGen, stage 1: predicted, linear fit	40,852	0.365	0.533	1.252	64.8 %	91.0 %
DIALibGen, stage 2: predicted, LOWESS	40,852	0.248	0.374	0.867	80.2 %	96.8 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes)	40,852	0.084	0.134	0.357	97.7 %	99.6 %
DIALibGen, stage 4: tuned, LOWESS	40,852	0.081	0.128	0.337	98.2 %	99.9 %
DIA-NN 2.0, stage 1: predicted, linear fit	40,852	0.349	0.522	1.132	66.0 %	92.6 %
DIA-NN 2.0, stage 2: predicted, LOWESS	40,852	0.294	0.441	0.943	73.7 %	96.1 %
DIA-NN 2.0, stage 3: tuned, as stored (run-01 minutes)	40,852	0.157	0.240	0.597	92.2 %	98.9 %
DIA-NN 2.0, stage 4: tuned, LOWESS	40,852	0.143	0.216	0.499	95.0 %	99.5 %
stage 5: refined, as stored (identified in run 01)	33,165	0.011	0.024	0.133	99.2 %	99.7 %
DIALibGen, stage 2: predicted, LOWESS, same precursors	33,165	0.238	0.359	0.825	81.7 %	97.3 %
DIALibGen, stage 3: tuned, as stored (run-01 minutes), same precursors	33,165	0.081	0.128	0.336	98.0 %	99.7 %
DIALibGen, stage 4: tuned, LOWESS, same precursors	33,165	0.078	0.123	0.319	98.5 %	99.9 %
stage 6: not identified in run 01, tuned, as stored	7,687	0.103	0.159	0.443	96.3 %	99.2 %
stage 6: not identified in run 01, tuned, LOWESS	7,687	0.100	0.156	0.419	96.8 %	99.7 %
stage 6: all precursors, as stored	40,852	0.023	0.027	0.230	98.7 %	99.6 %

The per-stage values are computed by `scripts/rt_stages.py` from the per-precursor table of the RT comparison and the refined run-01 library; `data/rt_stages.tsv` holds every row, including the medians of ΔRT (bias).